

DEVELOPMENT OF AN INTEGRATED MARKET FORECASTING MODEL FOR SHIPPING, AND SHIPBUILDING PARAMETERS

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SUMMARY

In a competitive world the utilization of resources is very critical. It is essential for a shipyard or a design agency to explore the market to know where and how the demand is likely to develop and behave. In this work, we present a forecasting model for an integrated marketing management model in which the shipyards are integrated at different levels (i.e. national or international) via consortiums. The proposed forecasting model uses a combined methodology driven by neural networks, weighted fuzzy logics and genetic algorithms, and the proposed integrated forecasting model has been used to simulate the shipping and shipbuilding markets. In the present forecasting model, neural networks (NN) are used to predict economic conditions at some future desired span of time (i.e. in our applications five years). In neural networks the back propagation neural networks are utilized. The output from the neural network model is input to a Weighted Fuzzy Petri Net based decision modeler (WFPNBDM). Another input to the WFPNBDM is from the expert's analysis (i.e. from organizations like International Monetary Fund, World Bank, and United Nations) dealing with the impact analysis of nanoeconomic events, geo-political environment, and weather conditions at future date. The WFPNBDM is designed using fundamentals from the theory of Petri nets, and genetic algorithms (GA). Finally, three case studies (i.e. global container traffic, new shipbuilding price indices, and market shares of shipbuilding nations) driven by the real world data and applications are presented to show the applicability of the proposed model, and ideas.

1. INTRODUCTION

Shipbuilding is an important, and strategic industry. Since, ships are complex and complicated products which are mainly manufactured with steel plates and assembly of outfit items and machinery. Thus a healthy shipbuilding industry also pushes the demand of materials and items, and hence creating a network of ancillary industries revolving around it, [1]. Utilizing this many countries have used shipbuilding to build their industrial infrastructures. Initially UK used the shipbuilding to develop and maintain not only its territorial but industrial empire also; and later Japan had used shipbuilding in the 1950's and 1960's to rebuild its industrial infrastructure of mainly manufacturing based industries. South Korea had used shipbuilding as a strategic industry to develop its economy around manufacturing based large industrial groups; and currently China is in the process to repeat these successful models with large governmental investments to develop its manufacturing base. Slowly other developing nations like Vietnam are trying to build their shipbuilding capabilities. Similarly, other European nations like Poland, Denmark, and Spain are either trying to re-enter into the world of shipbuilding or are slowly thinking to support and develop their existing shipbuilding industry.

Generally, the products of the shipbuilding industry are amongst the most technologically complex and sophisticated manufactured by any industry. The modern process of shipbuilding is also extremely complicated and sophisticated, and modern shipyards are highly efficient industrial enterprises which require extremely innovative business practices, and competent management. Since shipbuilding is an important industry for nations which are trying to develop their industrial

base, nations show a very strong tendency towards directly or indirectly supporting their shipbuilding industries; and because of this shipbuilding industry is fiercely competitive and utilization of resources becomes very critical.

Therefore, to remain competitive and cost efficient it is essential for a shipyard or design agency to explore the market to know where and how the demand is likely to develop and behave in future. The identification of emerging shipbuilding markets, and the forecasting of the shipping and shipbuilding parameters are complex and complicated processes because some parameters like weather, economy, geo-political situations which influence shipping and shipbuilding are inherently non-linear, and extremely volatile in nature. Because of this uncertainty in prediction of shipping and shipbuilding parameters, there exists two conflicting opinions, and these are as follows:

- a. The shipbuilding industry has a cyclic nature of business, and because of this corrections do occur at different stages. This makes forecasting risky. This opinion is largely shared in the industry, and consulting firms, [2].
- b. Even after assuming that the shipbuilding industry is of complex nature and has a cyclic nature, we can explore simulation of the markets with modern soft computing tools. And, with significant research into the simulation of shipping and shipbuilding parameters, not only we can get the trends right but also the exact behavior over the short span in future can be predicted. This opinion is largely shared in the academics, and classification societies, [3-8].

We agree with the second opinion mentioned above. In this work, we present a forecasting model for an

integrated marketing management model in which the shipyards are integrated at different levels (i.e. national or international) via consortiums. We present a forecasting model to simulate the shipping and shipbuilding markets (i.e. global container traffic, and new shipbuilding price indices). In the forecasting model, a combined methodology driven by neural networks, Weighted Fuzzy Petri Net based decision modeler, and genetic algorithms has been used.

The remaining of this paper is organized as follows: The second section briefly discusses the integrated forecasting model; the third section presents three case studies that are used to simulate the markets; and the fourth section concludes the paper by identifying some future applications, and scope of research. Since the marketing management studies are conducted using case studies, case studies that are motivated by real world applications and based on real data, have been used in this work. Some technical details have not been included in this work to keep the paper within reasonable length. Comprehensive theoretical details and a thorough treatment of all the case studies of this paper can be found in [9].

2. AN INTEGRATED FORECASTING MODEL

2.1 INTEGRATED NUCLEUS DRIVEN CONSORTIUM OF SHIPYARDS

It is well known that integration based upon either 'process' or 'product', brings down the cost and improves the bottom-line profit for any industry, [1]. Hence, the individual shipyard is viewed as a unit in an integrated network of shipyards, and the design, manufacturing, and marketing model is driven and modeled around the nucleus (i.e. shipyard). In the integrated network of shipyards there are two possibilities such as,

- i. Integration and networking of shipyards at the national level: In this the shipyards are integrated and networked to form a consortium. In the consortium there is one main lead contractor shipyard, and then there are several smaller shipyards or companies; and they are integrated and networked for manufacturing process out sourcing from the lead contractor shipyard to the smaller shipyards or companies.
- ii. Integration and networking of consortiums at the global level: In this the consortiums of shipyards at different national level or countries collaborate. They are integrated and networked via strategic alliances and partnerships. The collaborative strategy is implemented through the creation of a joint company with representations from all the consortiums of shipyards.

A detailed description about the integrated nucleus driven consortium of shipyards can be found in [7-8].

2.2 BASIC DESCRIPTION OF THE MODEL

In a competitive world it is essential for an industry to design its future plans according to the emergence of the markets. We present a method in which a combined methodology driven by neural networks, fuzzy logics, and genetic algorithms (Haykin [10], Ross [11], and Chambers [12]) is used to develop a forecasting model for shipbuilding and shipping market. We use neural networks (NN) to predict economic conditions at some future desired span of time (i.e. in our applications five years). In neural networks the back propagation neural networks are utilized. The output from the neural network model is input to a Weighted Fuzzy Petri Net based decision modeler (WFPNBDM). Another input to the WFPNBDM is from the expert's analysis (i.e. from organizations like Asian Development Bank, International Monetary Fund, World Bank, and United Nations, etc.) dealing with the impact analysis of nanoeconomic events, geo-political environment, and weather conditions at a future date. The WFPNBDM is designed using fundamentals from the theory of Petri nets, and genetic algorithms (GA). The present forecasting model is shown in Fig. 1.

To briefly show how the present forecasting model works; let us assume that at time t we want to predict some parameter from either shipping or shipbuilding at some future time $t+i+j$. This time period (i.e. $t+i+j-t$) is user specified, dependent upon the time structure of the systems, and can be day, week, year, decade and so on. The desired and relevant data is observed at time t , then the neural networks produce the outputs $p_{t+i}^u, p_{t+i}^v, \dots, p_{t+i}^w$, which are the inputs to the WFPNBDM. The other input to the WFPNBDM are inputs from expert's analysis at $t+i$ (i.e. for example the growth in container traffic is non-linearly related to the GDP growth; this non-linear relationship is modeled with neural networks; but the rate and pattern of GDP growth are best predicted by experts).

The WFPNBDM contains a base of decision rules in a linguistic manner, and these are called fuzzy rules. The inputs to the WFPNBDM are translated into linguistic terms via a procedure of fuzzification, and then these are processed using the fuzzy rules, fuzzy logic inference. In the next step, via a defuzzification and firing procedure a quantitative prediction of the selected shipping or shipbuilding parameter is made at time $t+i+j$. We investigate the forecasting of few chosen shipping, and shipbuilding parameters for example for shipping: global container traffic, and for shipbuilding: new shipbuilding price indices, and market shares of shipbuilding nations.

As shown in the Fig. 1, the starting point is a series of neural networks that are used to forecast each of the economic variables in the chosen market. These networks are indicated as $NN_u - NN_v - NN_w$ where $u+v+w$ is the number of economic variable that are

selected for the chosen market. Each of these neural networks has a number of inputs $X_{(1,...,n_U, or n_V, or n_W),t}^{u, or v, or w}$ which would decide the structure of the neural networks. The inputs to the networks are at a time t , and they correspond to the values of those variables at time t on which the market economic variables for chosen parameter depend. The output of these networks (i.e. $p_{t+i}^{u, or v, or w}$) corresponds to the values of the economic variables at time $t+i$ where i refers to the future period of time for which the forecasting is done. These serve as inputs to the WFPNBDM along with inputs from analysis of the experts. The WFPNBDM follows an algorithm which then results in the final values of economic variables.

In this work, we investigate shipping and shipbuilding markets, and we present the model parameters for new shipbuilding price indices. The details on other model parameters for global container traffic, and market shares of shipbuilding nations can be found in our other work [9]. We use the economic functional variables as,

- i. Ship supply (Specific ship type) - SSF_t ,
- ii. Ship demand (Specific ship type) - SSD_t ,
- iii. New shipbuilding price (Specific ship type) - $NSBP_{P_t}$,
- iv. Ship scrapping price (Specific ship type) - $SSCP_{P_t}$, and,
- v. Ship second hand price (Specific ship type) - $SSEP_{P_t}$.

This implies that in our forecasting model there are five neural networks, and each of them corresponds to each of the above mentioned economic functional variables. The subscript ' t ' refers to the value of these variables at time t .

We use four user specified ranges for the new ship building price indices, and these are as: Bulker – bulk carrier ships; Tanker – oil tanker ships; Container – 500 – 9500 TEU container ships; and OCS – other carrier ships. It means that each neural network will have four sub-neural networks in the model. Each of these sub-neural networks will have some economic functional variables as inputs. The sub-neural networks are shown in Fig. 2, and their inputs and corresponding outputs are given in Table 1. We use the past data (i.e. for different variables this varies from last 1 year to 32 years) for the training of neural networks. Since, for some data there is a large non-linearity and fluctuations we use 12-months moving averages that are generated from the monthly data, and use these data sets in the integrated forecasting system.

The sub-neural network learns the non-linearity of the system from data set, and forecast is made on this learnt non-linearity. We select the inputs of the economic

functional variables to the sub-neural networks at time $t-1$ as,

- i. Ship supply = SSF_{t-1} ,
- ii. Ship demand = SSD_{t-1} ,
- iii. New shipbuilding price - in million \$ US = $NSBP_{t-1}$,
- iv. Ship scrapping price - in 000 \$ US/Capacity = $SSCP_{P_{t-1}}$,
- v. Ship second hand price - in 000 \$ US/ Capacity = $SSEP_{P_{t-1}}$,
- vi. Steel price – in \$/metric tonne = $SP_{P_{t-1}}$,
- vii. Crude oil price – in \$ US per barrel = $O_{P_{t-1}}$,
- viii. Crude oil consumption - in 000 barrels per day = $O_{C_{t-1}}$,
- ix. Global interest rate - $GISinr_{t-1}$,
- x. Global inflation rate - $GISifr_{t-1}$, and,
- xi. Exchange rate (currency rate versus international currencies: Euro, and \$ US) - EGR_{t-1} .

Table 1: Inputs and outputs of sub-neural networks

Sub-neural network	Inputs	Outputs
1. Ship supply	SSF_{t-1} , $NSBP_{t-1}$, $SSCP_{t-1}$, $SP_{P_{t-1}}$, and EGR_{t-1}	SSF_t
2. Ship demand	SSD_{t-1} , SSF_{t-1} , $O_{P_{t-1}}$, $O_{C_{t-1}}$, $NSBP_{t-1}$, and EGR_{t-1}	SSD_t
3. New shipbuilding price	$NSBP_{P_{t-1}}$, SSF_{t-1} , SSD_{t-1} , $SP_{P_{t-1}}$, $GISinr_{t-1}$, $GISifr_{t-1}$, and EGR_{t-1}	$NSBP_{P_t}$
4. Ship scrapping price	$SSCP_{P_{t-1}}$, SSF_{t-1} , SSD_{t-1} , $GISinr_{t-1}$, $GISifr_{t-1}$, and EGR_{t-1}	$SSCP_{P_t}$
5. Ship second hand price	$SSEP_{P_{t-1}}$, SSF_{t-1} , SSD_{t-1} , $GISinr_{t-1}$, $GISifr_{t-1}$, and EGR_{t-1}	$SSEP_{P_t}$

Through these via a process of fuzzification, and defuzzification and firing the inputs lead to the final outputs. In the first layer there are three WFPNBDMs and these are used to predict the final values of the first stage decision variables at time frame ' $t+1$ '. In the second layer there is one WFPNBDM and it utilizes the inputs from the three first layer WFPNBDMs and predicts the final desired second stage decision variables (i.e. ship supply SSF_{t+2} , ship demand SSD_{t+2} , new shipbuilding price $NSBP_{t+2}$, new ship order SNO_{t+2} , and new shipbuilding price index $NSBPI_{t+2}$), at time frame ' $t+2$ '.

Inputs for economic parameters

$x^u_{I,t}$, $x^u_{2,t}$, $x^u_{n_u,t}$ → Neural network (NN_u) → $p^{u,t+i}$

$x^v_{I,t}$, $x^v_{2,t}$, $x^v_{n_v,t}$ → Neural network (NN_v) → $p^{v,t+i}$

$x^{wI}_{I,t}$, $x^{wI}_{2,t}$, $x^{wI}_{n_{wI},t}$, $x^{w2}_{I,t}$, $x^{w2}_{n_{w2},t}$, $x^{w3}_{I,t}$, $x^{w3}_{n_{w3},t}$ → Neural network (NN_{w1}) → $p^{wI,t+i}$

→ Neural network (NN_{w2}) → $p^{w2,t+i}$

→ Neural network (NN_{w3}) → $p^{w3,t+i}$

Prediction of economic conditions

Crisp input for the Weighted Fuzzy Petri Net based decision modeler

Input from expert's analysis (i.e. from organizations like IMF, WB, and UN) for the following:

- impact analysis of nanoeconomic events, geopolitical environment, and weather conditions at time $t+i$.

Weighted Fuzzy Petri Net based decision modeler (Two layers)

Fuzzification (Activated Fuzzy sets and activated truth value) → **Fuzzy rule bank** (Activated Fuzzy rules) → **Fuzzy inference** → **D and F process** (Clipped Fuzzy membership functions)

Crisp output from the Weighted Fuzzy Petri Net based decision modeler

Market decisions y_{t+i+j}

Basic nomenclature

- $x^{u, \text{or } v, \text{ or } w}_{(1, \dots, n_u, \text{ or } n_v, \text{ or } n_w)_t}$ = input parameters to neural networks at time t .
- $p^{u, \text{or } v, \text{ or } w}_{t+i}$ = prediction of neural networks at time $t+i$.
- y_{t+i+j} = output produced by WFPNBDM at time $t+i+j$.
- D and F process: Defuzzification, and firing process.

The inputs and outputs of first layer WFPNBDMs, and second layer WFPNBDM are listed in Table 2. The complete integrated forecasting system is shown in Fig. 3.

2.2 A BRIEF DESCRIPTION OF NEURAL NETWORKS, AND WEIGHTED FUZZY PETRI NET BASED DECISION MODELLING

Normally, a neural network (i.e. precisely artificial neural network) is a layered, and hierarchical structure, and it consist of input layers, output layers, and hidden layers which are located between these input and output layers. Each of these layers has a number of processing elements that are called neurons (or nodes/units). The neurons are interconnected via signal paths with weights; receive their inputs either from outside of the network via neurons in the input layers, or from other neurons via neurons in the hidden layers or output layers. After receiving the inputs the neurons compute their outputs by their transfer or activation functions, and these outputs are sent either to other neurons in the networks or to the output of the network layers as per the demands of network structure.

We use feedforward neural networks that are trained by the supervised learning back propagation algorithm. In the feedforward neural network, we use the ‘gradient descent’ technique ([10], and [13]) to optimize the weights of the neural network in a backward, and layer-by-layer manner so that the vector of the neural network outputs for a vector of training inputs achieves the desired outputs. The feedforward neural network is shown in Fig. 4 ([3], [5-6], [10], and [14]).

It has been observed that more complex a system becomes traditional mathematical methods are less able to model it; whereas humans can describe and deal with a complex situation with more ease using natural language and imprecise thinking [15]. This ability of natural language and imprecise thinking motivated the ‘fuzzy sets’, and ‘fuzzy logic’, [15]. As per fuzzy set theory an element can belong to set 1 and set $not - 1$ at the same time, and this allows ambivalence in fuzzy set theory. Fuzzy set theory is a response to the insufficiency of Boolean algebra in which only ‘0’ and ‘1’ are recognized. In fuzzy set theory everything is a matter of degree. Fuzzy logic is a set of reasoning with fuzzy sets, fuzzy truths, fuzzy connectives, and fuzzy rules of inference, [11], and [15].

Fuzzy sets tries to emulate human reasoning in a natural, logical, systematic, and mathematical way. Therefore, they are able to mimic human thinking and decisions very well, [11]. In the world of business, most of the information that is essential for decision making is imprecise, and ambiguous (i.e. for example market penetration possibility is expressed as likely, very likely, and unlikely, etc.). As any other competitive business, the shipping, and shipbuilding markets are full with fuzzy information, and fuzzy reasoning. It is appropriate

to use fuzzy sets whenever we have to deal with ambiguity, vagueness and ambivalence in mathematical or conceptual models of empirical phenomena, [11].

Fuzzy logic models linguistic terms mathematically, and maps the numerical (i.e. crisp) onto verbal (i.e. fuzzy), and vice versa. Therefore it builds a bridge between the two sets of completely different domains or reach (i.e. world). Because of this fuzzy logic is better suited to model and analyze economic and social systems. A fuzzy logic based system involves four major operations: fuzzification, activation of fuzzy rules, fuzzy inference, and defuzzification and firing process. The schematic of the fuzzy system is shown in Fig. 3. Normally, the inputs and output variables of the fuzzy system are called fuzzy variables, and the collection of all sets to which a fuzzy variable can belong is set called the universe of discourse. The universe of discourse is covered by overlapping fuzzy sets that are defined by fuzzy membership (i.e. truth value) functions. The degree of membership (i.e. truth value) that an element belongs to a certain fuzzy set is determined by the corresponding fuzzy membership function. In general fuzzy sets are marked with intrinsic linguistic terms or words. The crisp numerical inputs are converted into linguistic terms (i.e. fuzzy sets) along with their truth values based on the fuzzy membership functions via fuzzification. In our work, the defined linguistic levels for decision variables are listed in Table 3. We use trapezoidal fuzzy membership functions for decision variables (i.e. SSD , SSF , $NSBP$, and SNO). The fuzzy rule bank is a collection of fuzzy rules which contain a series of linguistic expressions. Normally, fuzzy rules are in the form of *IF – THEN* statements that interpret the linguistic input information obtained from the process of fuzzification, and then provide linguistic output information. Following Chen [16], and assuming that there is a fuzzy production rule R_i appearing in the knowledge base of a rule based system which is shown as follows:

$$R_i : IF d_j \text{ and } d_k THEN (CF \overline{\mu_i}), \text{ and} \quad (1)$$

$$IF X_1 \text{ is } A_j \text{ AND } X_m \text{ is } A_m THEN Y \text{ is } C, \quad (2)$$

where d_j ($X_1 \text{ is } A_j$), and d_k ($X_m \text{ is } A_m$) are propositions, $\overline{\mu_i}$ is a fuzzy number defined in the universe of discourse $[0,1]$ indicating the certainty factor value of the rule R_i , and the weights of the propositions d_j , and d_k are $\hat{w}_j$ and $\hat{w}_k$ respectively. Additionally, in Equations (1) and (2) $\hat{w}_j$ and $\hat{w}_k$ are fuzzy numbers defined in the universe of discourse $[0,1]$, and A_j and C are fuzzy sets (i.e. fuzzy values). Assuming that the fuzzy truth value of the proposition d_j is $\overline{tv_j}$, where $\overline{tv_j}$ is a fuzzy number defined in the universe of discourse $[0,1]$. Then, the fuzzy truth value of the proposition d_k is equal to $\overline{tv_j} \otimes \overline{\mu_i}$, where “ $\otimes$ ” is the multiplication operator of fuzzy numbers.

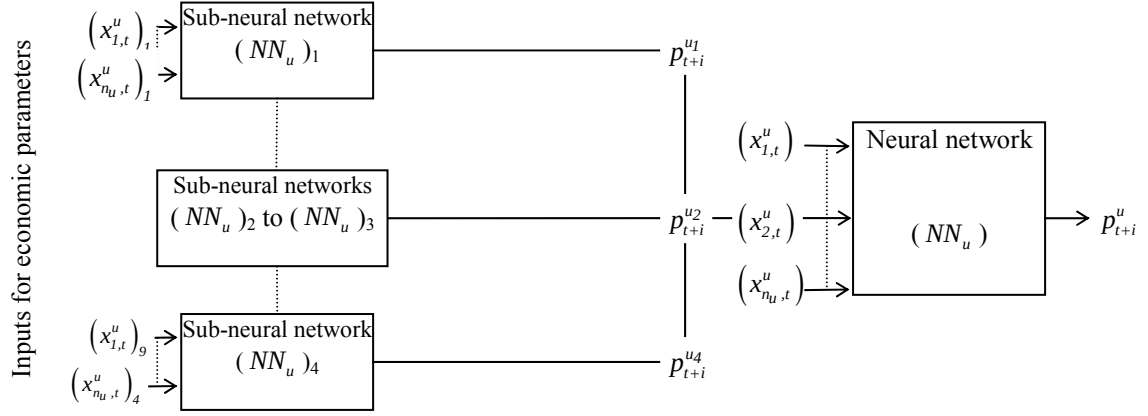


Figure 2: The structure of the sub-neural networks in the integrated forecasting system

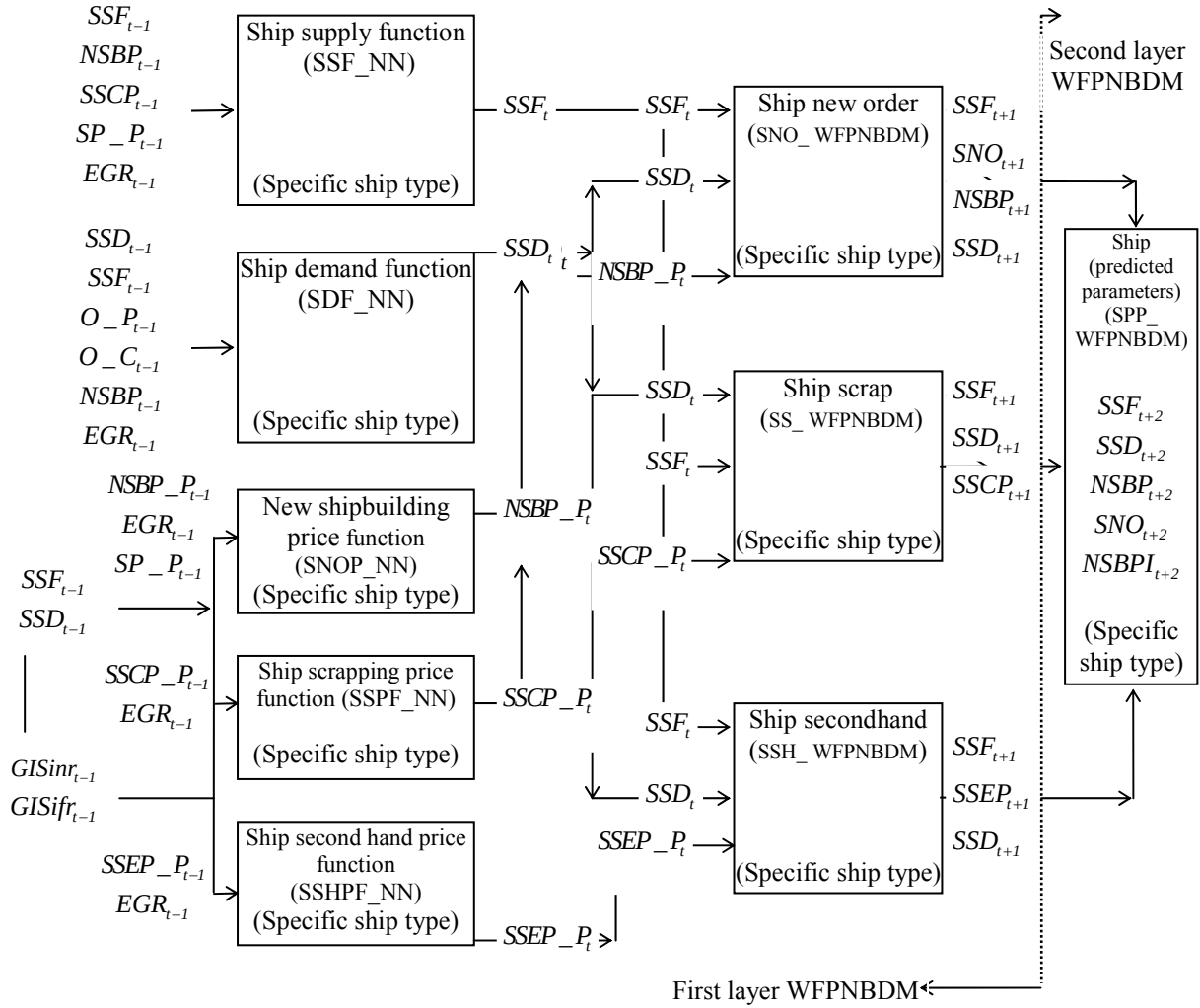


Figure 3: The architecture of integrated forecasting system

Equations (1) and (2) are very basic and all other types of fuzzy rules (i.e. for example rules involving *OR*, *NOT*, *LOGICAL OPERATORS* like *IF – BUT – THEN*, and multiple consequents, etc.) can be re-converted into the forms of Equation (1) and (2).

The fuzzy rule bank is expressed as a linguistic matrix for numerical implementation. If the linguistic input information is available then the fuzzy rules are determined as per the linguistic fuzzy rule matrix.

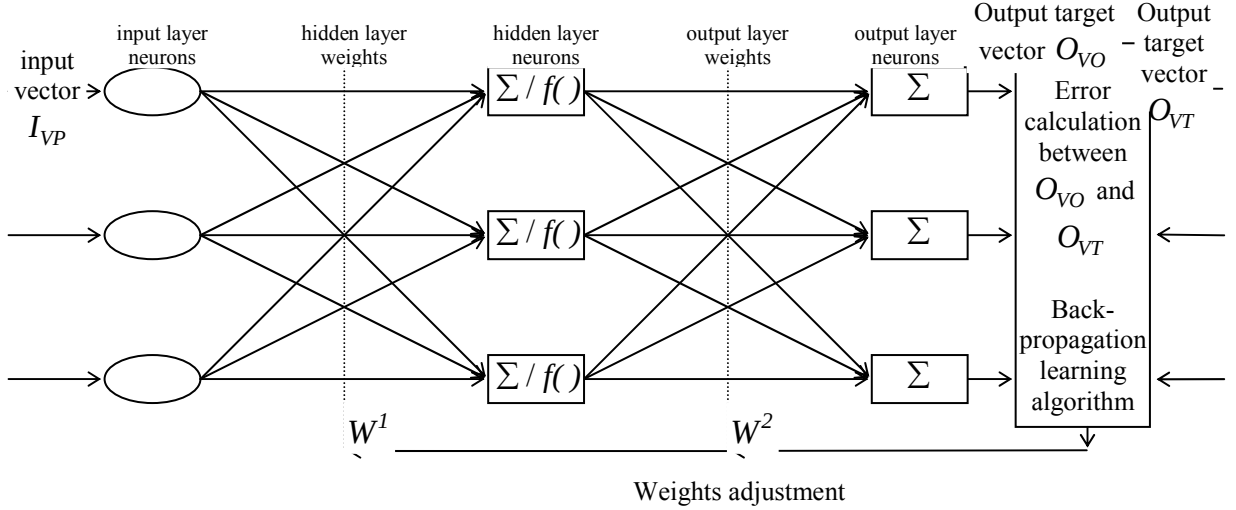


Figure 4: The feedforward neural network

In our applications we assume four different basic fuzzy rules for decision variables (i.e. SSD , SSF , $NSBP$, and SNO). The rules are,

RULE1 : IF $NSBP_t$ is SL and SSF_t is SH and SSD_t is SH , THEN SNO_{t+1} is L , (3)

RULE2 : IF $NSBP_t$ is SL and SSF_t is SH and SSD_t is N , THEN SNO_{t+1} is L , (4)

RULE3 : IF $NSBP_t$ is N and CSF_t is SH and CSD_t is SH , THEN SNO_{t+1} is SL , and (5)

RULE4 : IF $NSBP_t$ is N and SSF_t is SH and SSD_t is H , THEN SNO_{t+1} is SH . (6)

The logical and numerical operations are implemented on the activated fuzzy rules via the process of inference, to determine their levels of activations. The levels of activations are determined according to the truth values of the fuzzy sets bounded by the activated fuzzy rules, and result into clipped output fuzzy sets. In this work, we use ‘correlation-minimum inference’ (i.e. simply cutting off the top of the fuzzy set, and using only that area of the set which lies below the user specified current degree of belief (DoB), [11], [17], and [18]). The low belief in a fuzzy concept implies that the true discrete value lies outside the supremum region. The supremum region expands as belief decreases from maximum to minimum (i.e. for implementation $1 - 0$) until it equals the support of the fuzzy set. Since, in our applications, we wish to retain the impact of a set even if its degree of belief (DoB) is decreasing, and assume that the percent area of the set is notably higher than its DoB always; therefore correlation-minimum inferencing is better suited for our applications. The output fuzzy sets (i.e. clipped fuzzy sets) are demystified and converted into the crisp and numerical outputs via the process of defuzzification and firing. In this work, we use ‘area-centroid weighting’ (i.e. simply cutting off the top of the fuzzy set, and using only

that area of the set which lies below the user specified current degree of belief (DoB), [11], and [18]) for defuzzification and firing. In our applications we use symmetrical fuzzy sets (i.e. trapezoids), and the centroid position is independent of DoB and inferencing Method. The centroid is always located at the mid-point of the supremum range. And, this allows the supremum position (i.e. average position) to be used instead of calculating the centroid on each cycle through the controller which is computationally expensive. Furthermore, we consider the fuzzy set as a singleton. After deciding the input and output variables of the system to be modeled the Weighted Fuzzy Petri Net based decision modeler (WFPNBDM) is designed.

Following, Chen [16], and using the fundamentals of Petri nets [19], and fuzzy Petri nets [20-23], we use a generalized Weighted Fuzzy Petri Net structure (WFPNS), as,

$$WFPN = (P, T, D, I, O, f, \alpha, \beta, W), \quad (7)$$

where,

$$P = \{p_1, \dots, p_n\} \text{ is a finite set of variables,} \quad (8)$$

$$T = \{t_1, \dots, t_m\} \text{ is a finite set of transitions,} \quad (9)$$

$$D = \{d_1, \dots, d_n\} \text{ is a finite set of propositions,} \quad (10)$$

$$P \cap T \cap D = \emptyset, |P| = |D|, \quad (11)$$

$$I : T \rightarrow P^\infty \text{ is the input function, a mapping from transitions to sets of variables,} \quad (12)$$

$$O : T \rightarrow P^\infty \text{ is the output function, a mapping from transitions to sets of variables,} \quad (13)$$

$$f : T \rightarrow \text{fuzzy numbers in } [0,1] \text{ is an association function, a mapping from transitions to fuzzy numbers defined in the universe of discourse } [0,1], \quad (14)$$

$$\alpha : T \rightarrow \text{fuzzy numbers in } [0,1] \text{ is an association function, a mapping from variables to fuzzy numbers defined in the universe of discourse } [0,1], \quad (15)$$

$\beta : P \rightarrow D$ fuzzy numbers in $[0,1]$ is an association function, a bijective mapping from variables to propositions, (16)

$W : P \rightarrow$ fuzzy numbers is an association function, a mapping from variables to fuzzy numbers. (17)

The design of WFPNBDM requires that the number of fuzzy sets, their associated membership functions for the input and output variables, transitions for the input and output variables, associated weights for the input and output variables, and the specifications of the fuzzy rules are determined efficiently. The WFPNBDM is designed so that it can model the system efficiently. The complexity of a WFPNBDM increases with the unknown parameters.

As mentioned previously, we use two layers of WFPNBDMs. It means that in the first layer there are three WFPNBDMs for decision variables (i.e. ship new order SNO , ship scrap price $SSCP$, and ship secondhand price $SSEP$), and in second layer there is one WFPNBDM for desired second stage decision variables (i.e. ship supply SSF , ship demand SSD , new shipbuilding price $NSBP$, ship new order SNO , and new shipbuilding price index $NSBPI$). These, are designated: first layer - $SNO_WFPNBDM$, $SS_WFPNBDM$, and $SSH_WFPNBDM$; second layer - $SPP_WFPNBDM$. The mapping relationships for both layers WFPNBDMs are shown in Fig. 3. The transition, and relative weights of different variable functions are listed in Table 4.

Table 4: Transition, and relative weights of different variable functions

S. no.	Transition	Relative weights
First layer WFPNBDM	$SSCP \rightarrow$ $\rightarrow SSEP \rightarrow$ $\rightarrow SNO$	$SSCP = 1$, $SSEP = 0.875$, $SNO = 0.75$
Second layer WFPNBDM	$SSD \rightarrow SSF$ $\rightarrow NSBP \rightarrow$ $\rightarrow SNO \rightarrow$ $\rightarrow NSBPI$	$SSD = 1$, $SSF =$ 0.875 , $NSBP =$ 0.85 , $SNO =$ 0.85 , $NSBPI =$ 0.75
$\rightarrow$: mapping.		

GAs do not require that the objective function is continuous, twice differentiable, and mathematically well formulated. Also, GAs operate on a domain (i.e. population) of potential solutions (i.e. individuals, or chromosomes) at each step of iteration and do not evolve a single solution. Additionally, GAs take a pre-conceived random search strategy and ‘a priori’ knowledge about the searching space where the solution will lie is not required. Since, GAs operate on a domain of potential solutions and search randomly, so they may miss the local optimal solutions, [12]. Because, of this limitation GAs are better suited for higher dimensional, ill defined, and nonlinear problems. To automate the simultaneous

identification of both the fuzzy membership functions and fuzzy rules, we use genetic algorithm (GA) driven optimization. The details of the algorithm can be found in [9], [12], and [24].

It is known that the nanoeconomic parameters (i.e. weather nep_1 , natural calamities nep_2 , GDP growth nep_3 , stability in exchange rates nep_4 , and geo-strategic-political climate in middle east Asia nep_5) affect almost every shipping and shipbuilding parameters (i.e. ship demand, ship supply, new shipbuilding price, and new ship order, etc.). The weighted fuzzy system allows for the inclusion of the effects of nanoeconomic parameters.

There can be many nanoeconomic parameters, and though the inclusion of each parameter is possible in the weighted fuzzy system, but to avoid more complexity and keep our modeling system simple, we select five specific parameters (i.e. weather – monsoon = nep_1 ; natural calamities – cyclone, and tsunami = nep_2 ; GDP growth nep_3 , exchange rates – exchange rate against \$ US = nep_4 ; and middle east stability – strategic and political environment = nep_5). For example, let us consider the Rule 1 as in Equation (3),

RULE1 : IF NSBP_t is SL and SSF_t is

SH and SSD_t is SH, THEN SNO_{t+1} is L

The occurrence of any nanoeconomic event will have a probability (p_b), and impact factor (ipf). It means that any nanoeconomic event will affect the decision variables (i.e. for example SNO_{t+1} will move to L or N or VH). We suppose that there is ‘ p_{b1} ’ probability and ‘ ipf_1 ’ impact factor of the nanoeconomic parameters nep_1 that SNO_{t+1} will shift to L ; ‘ p_{b2} ’ probability and ‘ ipf_2 ’ impact factor of the nanoeconomic parameters nep_2 that SNO_{t+1} will shift to L ; ‘ p_{b3} ’ probability and ‘ ipf_3 ’ impact factor of the nanoeconomic parameters nep_3 that SNO_{t+1} will shift to SH ; ‘ p_{b4} ’ probability and ‘ ipf_4 ’ impact factor of the nanoeconomic parameters nep_4 that SNO_{t+1} will shift to N ; and ‘ p_{b5} ’ probability and ‘ ipf_5 ’ impact factor of the nanoeconomic parameters nep_5 that SNO_{t+1} will shift to H . Then, we consider each of these cases individually, and we find the final inference fuzzy set and the centroid of the area under the weighted fuzzy set. The centroid under the weighted fuzzy set gives the crisp output for each of these cases. We compute the consolidated crisp output for the nanoeconomic parameters: let a_1^* , a_2^* , a_3^* , a_4^* , and a_5^* be the normalized areas; and cg_1 , cg_2 , cg_3 , cg_4 , and cg_5 be the centroid of area of the final inference sets (i.e. clipped sets); corresponding to each of the possibilities.

Then, desired crisp output,

$$= \frac{\left(p_{b1} \cdot ipf_1 \cdot a_1^* \cdot cg_1 + p_{b2} \cdot ipf_2 \cdot a_2^* \cdot cg_2 + p_{b3} \cdot ipf_3 \cdot a_3^* \cdot cg_3 \right. \\ \left. + p_{b4} \cdot ipf_4 \cdot a_4^* \cdot cg_4 + p_{b5} \cdot ipf_5 \cdot a_5^* \cdot cg_5 \right)}{\left(p_{b1} \cdot ipf_1 \cdot a_1^* + p_{b2} \cdot ipf_2 \cdot a_2^* + p_{b3} \cdot ipf_3 \cdot a_3^* + \right. \\ \left. p_{b4} \cdot ipf_4 \cdot a_4^* + p_{b5} \cdot ipf_5 \cdot a_5^* \right)} \quad (18)$$

and thus the nanoeconomic parameters are included in the fuzzy model.

The algorithms presented above have been implemented with Matlab[®] 7.1, and in C++ using its object-oriented features on a Silicon GraphicsTM OriginTM 200 workstation. The results are presented graphically in Ms-Excel^{###} 2000.

4. SIMULATION RESULTS AND DISCUSSION

In our case studies we have used real world data from various sources that include World Shipyard Monitor [25], KG Shipping and Finance [26], Shipping Intelligence Weekly [27], LNG Trade and Transport [28], China Intelligence Monthly [29], and Asian Development Bank [30].

4.1 SHIPBUILDING FORECASTING: NEW SHIPBUILDING PRICE INDICES

Since shipbuilding projects take significant time to be completed and actual costs may change until the delivery of the vessel, it is important for the shipyard to know in-advance the price behavior for the next couple of years. The price of a new ship at which the contract is signed is based upon forward assumptions regarding the material costs, interest rates, exchange rates, and other economic parameters. These assumptions are continuously reviewed and results are updated whenever new or better information is received. A shipyard can identify and explore new markets or can incorporate the price behavior in its contracting prices of new ships.

Case study 1. A shipyard in a developing country is planning to explore the market for different types of ships. In this regard, the shipyard approaches a design team/agency for market forecast on the new shipbuilding price indices.

Discussion: For container ships, we analyze the past market for the last ten years (i.e. 1997 – 2006), and using the model discussed previously in Fig. 1, the forecast is made for the next five years (i.e. 2007 – 2011). The forecasted new shipbuilding price index – container (NSBPI – C) is shown in Fig. 5.

From Fig. 5, it is clear that in 2001 – 2002 there was a significant fall in the new shipbuilding price index (container), and therefore ship prices at the end of 2002 were generally lower across the board than at the end of

2001. The prices recovered slightly in 2004, and then moved upwards though at a slow rate. As per our forecasted estimates this rise in ship prices is likely to continue till 2007. After, 2007, ship prices again are expected to fall marginally. In the period after 2001 – 2002, the \$ US has been declining continuously against the currencies (i.e. Euro, Won and Yen) of major shipbuilding nations, but this has not resulted in a price increase (i.e. in \$ US) across the board. The reasons for this are mainly the stiff competition and overcapacities amongst the Korean shipyards (i.e. our assumption is at 29.3%; and since the overcapacity is very high at Korean yards, they need to partially or completely fill their large building docks by trying to get any order for any ship that appears in the market, and assure enough cash flow to roll over their short terms debts).

Although, there is a strong demands (i.e. arising because of tonnage replacement, massive cargo requirements for domestic supplies in growing Asian economies like China, Japan, and India, and very large to ultra large container ships for economies of scale), this stiff competition and overcapacity largely compensate any possible upward movement of new shipbuilding price index (container) for long, and hence keep the new shipbuilding price index low. In this regard, in 2006 – 2011, though, a marginal decrease in ship prices is very much probable, but on a comparative scale this decrease is not significant. As a cautionary note, we can state that the new shipbuilding price index (container) is most likely to correct itself at around 2008 but overall outlook will remain steady and positive in the period 2006 – 2011. For the shipyard in the developing country, it means that if its current contract prices are competitive then with little cost re-structuring it can remain competitive and attract new orders for shipbuilding.

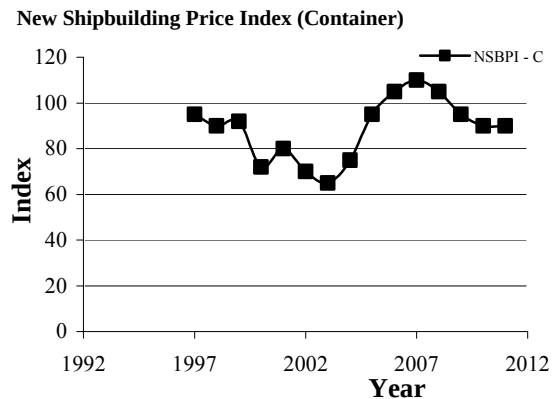


Figure 5: The forecasted new shipbuilding price index (container)

For tanker ships, we analyze the past market for the last twenty seven years (i.e. 1980 – 2006), and using the model discussed previously in Fig. 1, the forecast is made for the next five years (i.e. 2007 – 2011). The forecasted new shipbuilding price index – tanker (NSBPI

– T) is shown in Fig. 6. Tanker market is very sensitive and volatile, and there are always some periodic highs and lows. For example, the peak cost of an Aframax tanker was around 55 million \$ US in 1990, and its price fell to a low of 33 million \$ US in 1999. After, 1999 the prices started looking up, and at present an Aframax tanker will cost around 67 million \$ US. This recovery of prices in the tanker segment is likely to continue till 2008. After, 2008, ship prices again are expected to fall marginally. The overall tanker market looks quite firm because of some positive signs (i.e. arising from new demands because of expanding economies like China, and phasing out of tankers; and rising investment in tankers) in the coming two years. However after 2008 the tanker market is expected to correct itself because there will be an expected fall in the presently high freight rates, and the delivery of lots of new ships will cause additional pressure from the supply side. Also, though at present the demand has been growing very strongly as the world economy is surging ahead, but this is not sustainable in the long run.

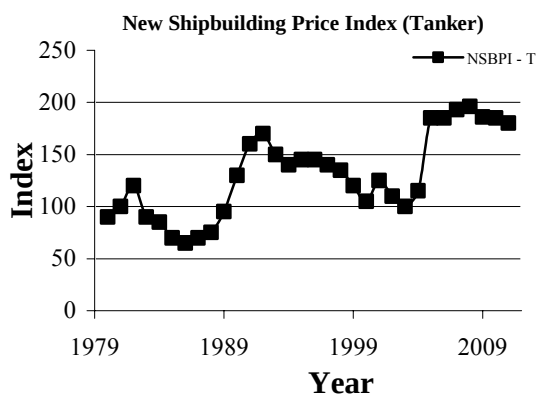


Figure 6: The forecasted new shipbuilding price index (tanker)

As previously, for bulkier ships, we analyze the past market for the last twenty seven years (i.e. 1980 – 2006), and using the model discussed previously in Fig. 1, the forecast is made for the next five years (i.e. 2007 – 2011). The forecasted new shipbuilding price index – bulkier (NSBPI – B) is shown in Fig. 7. Bulkier market is very similar to tanker market, and it too is sensitive and volatile. After, 2003 the prices started looking up, and this recovery of prices in the bulkier segment is likely to continue till 2008. After, 2008, ship prices again are expected to fall. The overall bulkier market looks firm because of some positive signs (i.e. rising investment in bulkiers by Japan, Germany, and Norway) in the coming two years. However, similar to the tanker market, after 2008 the market is expected to correct itself.

Overall, at present there is a positive feeling in the shipbuilding industry, and the market outlook is keeping the shipyard's order books full. This positivity is expected to continue in coming 2/3 years. However, as

we have shown there will be corrections in the markets, and in our opinion it is time for caution. As a summary of our analysis, we briefly present the positive signs, and also present the reasons for caution. The positive signs are,

- Arising demand out of lost tonnage: Around 1990, the old fleets were scrapped, and that has opened up a demand for tankers, and bulkiers. This shortage promoted an investment, and shipbuilding industry benefited from it.
- Firming of freight rates: Since, the demand was created in the cargo sector, it gave the positive push to the freight rates. And, recently freight rates have been high as compared in the past. This is likely to continue for another 2/3 years.
- Change in ship economics: Since, during 1990 – 2000, ship prices were normally low across the board, and this coupled with the low interest rates existing in modern economies made the new ships economical to acquire, and operate. These factors reduced the 'time charter equivalent cost' (i.e. for example for a new VLCC cost reduced from 45,500 \$ US per day in 1992 to 22,000 \$ US per day in 2004). This saving created a large investment in new shipbuilding. This is likely to continue for another 2/3 years.
- Stiff regulatory environment: The stiff and rigid regulations by the world bodies have discouraged old ships from the market. The tanker industry is expected to phase out all the single hull tankers, and this is and will create additional demand. This is a very positive outcome, and is likely to continue for another 2/3 years.
- Growth in Asia: Recently, in the last couple of years there has been strong growth in the Asian economies. And, this has created demands in the cargo, tanker, LPG/LNG, and container sectors. This has driven the cargo carrier rates very high because of strong demands of materials (i.e. iron ore, steel, and cement, etc.). Though, this is likely to continue for another 2/3 years, but after that it will slow down.

The reasons for caution are,

- De-firming of freight rates: Recently, freight rates have been on a record high. As big ships that have ordered in the past 2/3 years are expected to join service in coming future, the gap between demand and supply will be filled. After, this freight rates are expected to dip. This expected around 2008-2009.
- Firming of Asian currencies against \$ US: Recently, \$ US is getting de-firmed against all the major Asian currencies. This will make exports costlier for Asian countries, and will affect the demands and consumptions of developed nations adversely. This too is expected around 2008-2009.
- Firming of steel prices: Recently, international steel industry is getting integrated (i.e. via acquisition and mergers of major steel manufacturing units across

the globe), and this unitised integration is expected to make steel prices steady and firm in the coming future. It means that the material cost in shipbuilding will go up. This can happen around 2007-2008.

- o Slowing of growth in Asia: The recently strong growth in the Asian economies is largely expected to slow down. This will affect the cargo demand of Asia adversely. This can happen around 2008-2009.

Finally, we would like to conclude that the shipbuilding industry is having a good time at present, and this will continue for some time. However, as was mentioned previously shipbuilding is a very cyclical business, and a correction is expected in future. Shipyards need to tighten their production economies now, and after a caution at this time of prosperity down turns of future shall be overcome.

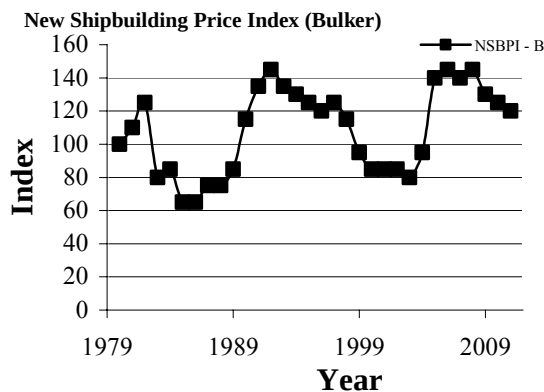


Figure 7: The forecasted new shipbuilding price index (bulker)

Case study 2. A manufacturing company in Asia is planning to enter into shipbuilding, and in this regard, the company approaches a consulting firm for market forecast on the global shipbuilding capacity shares.

Discussion: As previously, we analyze the past market for the last thirty two years (i.e. 1975 – 2006), and using the model discussed previously in Fig. 1, the forecast is made for the next five years (i.e. 2007 – 2011). The forecasted global shipbuilding capacity shares are shown in Fig. 8 (i.e. South Korea – SK; China – CHI; Europe – EUR; Japan – JAP; and others – OTH). As mentioned previously, the shipbuilding is an attractive industry for the nations which plan to develop their economies around manufacturing based industries. The successful examples of this approach include Japan, South Korea, and now China. The shipbuilding industry requires some strong infrastructure (i.e. reliable network of sub-contractors and vendors to supply materials and human resources, integrable supply chains of manufacturing, skilled workforce, expertise in basic sciences and engineering, infrastructure of logistics and connectivity) in a nation to succeed. This development of infrastructure requires heavy investment, and it is

normally provided by the governments in the initial phases. Because shipbuilding is important, in the 1950's and 1960's Japan had entered into shipbuilding to rebuild its industrial infrastructure. Japan was able to increase its share in Global Shipbuilding Capacity Shares (GSCS) to around 42.357 % (in % of CGT – cumulative gross tonnage). Later, South Korea has utilized the same philosophy and it had pursued shipbuilding as an industry integral to its economy. Also, recently in 2003 shipbuilding output exceeded the previous peak production levels achieved in the 1970s. In 2003, the output was 21.9 million CGT. And, in 2006, the deliveries were expected to cross 70 million dwt. In this buoyant period of shipbuilding, around 2002, South Korea consolidated its position as the leading shipbuilding nation with an output of 6.75 million CGT (i.e. exceeding Japan's 6.5 million CGT). Korea has maintained this lead and in 2006 its share was at 33.625% as against 31.125 % of Japan. Recently, shipbuilding has attracted a very high investment (i.e. GIBR - global investment by builder region, in 1996 = 34.1 billion \$US, in 2006 100.8 billion \$ US), and the order book has a current contract value of 275 billion \$ US. This huge investment has attracted the other Asian nations also in shipbuilding. And, in 1990s with large governmental investments, China started its entry into shipbuilding and pursued aggressively. It was able to gain a foothold, and slowly started to increase its share. The center of gravity of shipbuilding in the past has moved from Europe to Asia, and amongst Asian nations, it has moved from Japan to South Korea later. And, now though center of gravity is likely to remain in Asia but there will be an adjustment amongst the shares of Japan, Korea, and China. Japan and Korea both are likely to lose their shares, and China will gain in this process. Also, the share of others too is likely to increase at the expense of Europe, Korea, and Japan. Overall, the future outlook for Asian shipbuilding is positive, and some other nations too will be able to gain a foothold. The shipbuilding will remain competitive, and with new players like China, Vietnam, Spain, Denmark, and Norway, the competition will be fierce.

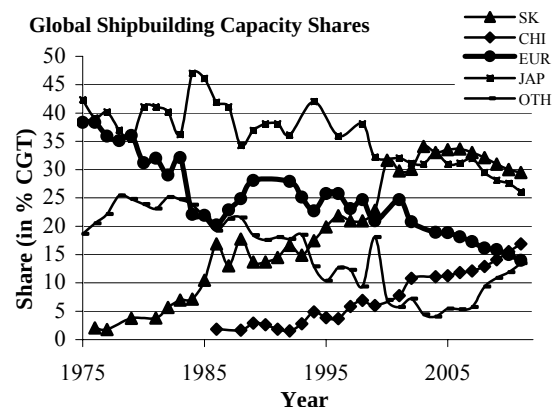


Figure 8: The forecasted global shipbuilding capacity shares

4.2 SHIPPING FORECASTING: GLOBAL CONTAINER TRAFFIC

Apart from academic interest or from concept design (i.e. for example though the concept design has been explored for container ships up to 18,000 TEU capacities but market forces overrule any detailed design beyond around 12,000 TEU capacities in near future 2010 - 2011); no new ship is designed till there is a demand of that particular type of ship. In this regard it is essential for a shipyard or a design agency to explore the market to know where and how the demand is likely to develop and behave. The estimation and forecasting of the demand for container ships or any other type of commercial ship is difficult and complex. Though, growth in the containerized transportation (i.e. container trade) or any other maritime transportation (trade) is basically driven by economic growth, but there are other factors that affect the volume of containerized transportation and these include exchange rate fluctuations, changes in the structure of economic policy, etc. It is difficult to forecast these variables (i.e. for example it is known that the container imports and exports are undoubtedly greatly affected by exchange rate movements, but the uncertainties involved in estimating exchange rates are immense). Though the world economy is considered resilient after the 1997 Asian currency crisis, the exchange growth rates during the coming decade (i.e. 2006 - 2016) are not expected to rise much further beyond the recent high rates. We assume that the average growth rate in the short term is similar to that of the recent past, and in the medium term is approximately equal to the long-term cumulative average growth rate for the world economy over the last 30 years. Furthermore, in the short run the developing countries in Asia are assumed to continue to keep their positive economic growth pattern. In our forecast we rely on the projection of International Monetary Fund (IMF) and extend the average growth rate from the period during which the IMF projections provided explicit forecasts to the remainder of the forecast period. Though, we do incorporate that there may be some good years or bad years within the forecasted period, but there will not be a major, prolonged economic slowdown on the scale of that of the early 1990s. Since, it is very difficult to predict the timing of downward economic cycles which can inevitably occur at any time, the horizon for our forecasts is medium term.

Case study 3. A shipyard in a developing country (i.e. India) is planning to explore the container ship market. In this regard, the shipyard approaches a design team/agency for market forecast on the global and Indian container traffic.

Discussion: We analyze the past market for the last twenty five years (UNR [31]), and using the model discussed previously in case study 1, the forecast is made for the next five years (i.e. 2006 – 2011). The forecasted

global container traffic is shown in Fig. 9, and listed in Table 5. As an input to the model in Fig. 1, we estimate volumes of imports and export from independent equations for individual countries in the Asia and the Pacific region (i.e. East Asia - China, Hong Kong, and Taiwan; North Asia - South Korea, Japan, North Korea, and Pacific Coast of the Russian Federation; South - East Asia - Brunei Darussalam, Cambodia, Indonesia, Malaysia, Myanmar, Philippines, Singapore, Thailand and Vietnam; and South Asia - Bangladesh, India, Pakistan, Sri Lanka, Iran, and Turkey), and for the countries outside of Asia and the Pacific region we estimate volumes of imports and export from separate equations for each 'region (i.e. a group of countries)'. In Table 5 and Fig. 9, we have shown the full origin-destination container volumes, and therefore we do not include empty containers. Furthermore, each container is counted only once during its entering journey, regardless of how many times it may be handled. As per our estimates in Table 5 and Fig. 9, the total number of full containers shipped internationally is expected to grow to 148 million TEU by 2011, up from 98 million TEU in 2004. But, the forecasted growth rate in the next few years (i.e. 2006 – 2011) is expected to be slower at the average rate of 6.7542 per cent per annum compared to 8.4345 per cent per annum during 1985 - 2005. These comparisons are clearly summarized in Table 5. The comparisons of our study's forecasts with those provided by private consulting firms (DSC [32], and Notteboom [33]) suggest that our global level estimates lie within the range of expert opinions, but slightly towards the more conservative and cautious end of that range. The cautionary and conservative trend in our estimates is because of assumptions that are made by us about future world economic growth.

5. CONCLUSIONS

This work has presented an integrated forecasting model (i.e. shipyards are integrated at different levels: national or international via consortiums) that is designed around a shipyard. This will make the supply and delivery of materials and items streamlined, sleek, and will allow horizontal integration amongst the different shipyards in the consortium. The forecasting model presented in this work uses a combined methodology driven by neural networks, Weighted Fuzzy Petri Net based logics, and genetic algorithms, and the integrated forecasting model has been used to simulate the shipping and shipbuilding markets. The application of neural networks allows the detection and learning of nonlinear relationships amongst the various shipping and shipbuilding decision variables and parameters. The application of Weighted Fuzzy Petri Net based logics and genetic algorithms allow the incorporation of nanoeconomic parameters (i.e. weather, natural calamities, GDP growth, stability in exchange rates, and geo-strategic-political climate in middle-east Asia). Hence, the shipping and shipbuilding market simulation with integrated forecasting model is representative of real world situations to a large degree.

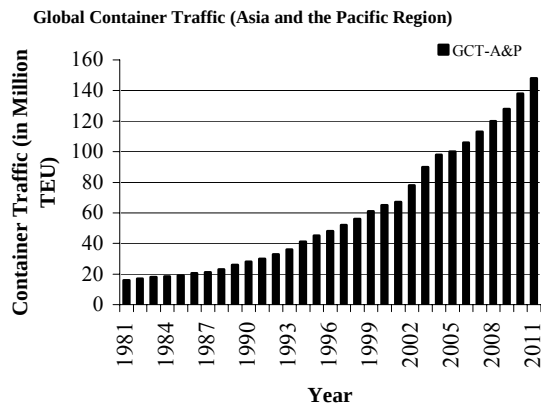


Figure 9: The forecasted global container traffic (Asia and the Pacific region)

Table 5: The past and forecasted values for global container traffic (Asia and the Pacific region)

S. no.	Past and forecasted global container volumes	
	Global container volume (in million TEU)	Growth rate on the previous year (in %)
1981	16	-
1982	17	06.250
1983	18	05.882
1984	18.5	02.778
1985	19	02.703
1986	20.5	07.895
1987	21.125	03.049
1988	23	08.876
1989	26	13.043
1990	28	07.692
1991	30	07.143
1992	33	10.000
1993	36	09.091
1994	41	13.889
1995	45	09.756
1996	48	06.667
1997	52	08.333
1998	56	07.692
1999	61	08.929
2000	65	06.557
2001	67	03.077
2002	78	16.418
2003	90	15.385
2004	98	08.889
2005	100*	02.041*
2006	106*	06.000*
2007	113*	06.604*
2008	120*	06.195*
2009	128*	06.667*
2010	138*	07.813*
2011	148*	07.246*
*Predicted values		

Since, the application of integrated forecasting model has been presented for the shipping and shipbuilding market parameters (i.e. global container traffic, new shipbuilding price indices, and market shares of shipbuilding nations) by simulating real world data and situations, the results of this work can be utilized by any shipyard to understand the market or enter into the market.

However, in this work we have neither attempted a generalized marketing model for any shipyard in any fiscal state, nor we have considered all the nanoeconomic parameters in the integrated forecasting model. A more detailed integrated forecasting model considering specific nanoeconomic parameters relevant to one particular nation can be an obvious extension of the present work. Also, a detailed sensitivity analysis of the integrated forecasting model incorporating different parametric values of decision variables has not been done. Each of the ideas presented in this work can be further explored 'in-detail' to develop a comprehensive long-term marketing management model for shipyards. Now, we shall be devoted to market our presented model amongst our industrial partners, and various governmental agencies to initiate our research into the development of a comprehensive forecasting model for shipping, and shipbuilding parameters. Our future work shall go in this direction, and currently this is under investigation.

6. ACKNOWLEDGEMENTS

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7. REFERENCES

1. STOPFORD, M., *Maritime Economics*, TF-ROUTL UK, 2nd edition, Dec. 1997, pp 397-402, 1997.
2. STOPFORD, M., *World Shipbuilding 2004*, Clarkson Research, UK, pp 1-5, 2004.
3. LI, J., AND PARSONS, M. G., Forecasting Tanker Freight Rate using Neural Networks, *Maritime Policy and Management*, Volume 24, no. 1, pp 9-30, 1997.
4. LI, J., AND PARSONS, M. G., The Use of Fuzzy Logic in Shipping and Shipbuilding Market Modeling, Analysis, and Forecasting, *Journal of Ship Production*, Volume 12, no. 2, pp 85-98, 1996.
5. LI, J., AND PARSONS, M. G., An Improved Method for Shipbuilding Market Modeling and Forecasting, *SNAME Transactions*, Volume 106, pp 157-183, 1998.
6. SHARMA, R., AND SHA, O. P., Development of an ERP Model for Modularly Designed Ships for Medium Scale Shipyards – II: Marketing Management, *Journal of Marine Engineering and Technology [Part A: Proceedings of the IMarEST]*, 2007, (To appear).
7. SHARMA, R., AND SHA, O. P., Development of an ERP Model for Modularly Designed Ships for Medium Scale Shipyards – I: Manufacturing Management, *Journal of Marine Engineering and Technology [Part A: Proceedings of the IMarEST]*, 2007, (To appear).
8. SHARMA, R., AND SHA, O. P., An Investigation into Modular Concept, and Strategic Alliances for

Design and Production of Naval Ships, *Defence Science Journal*, 2007, (Under review).

9. SHARMA, R., AND SHA, O. P., *An Analysis of Market for Feeder Vessels*, Technical Report TRP - 12/2007/2(2)/SDEE, Design Laboratory, Department of OE and NA, IIT, Kharagpur, India, pp 1–72, (2007).

10. HAYKIN, S., *Neural Networks: A Comprehensive Foundation*, 2nd edition, Prentice Hall, USA, pp 55–67, 1998.

11. ROSS, T. J., *Fuzzy Logic with Engineering Applications*, 2nd edition, John Wiley and Sons, USA, pp 132–147, 2004.

12. CHAMBERS, L. D., *The Practical Handbook of Genetic Algorithms: Applications*, 2nd edition, Chapman and Hall/CRC, USA, pp 67–78, 2000.

13. RUMELHART, D. E., HINTON, G. E., AND WILLIAMS, R. J., *Learning Internal Representations by Error Propagation, Parallel Distributed Processing, Volume I: Foundations*, D. E. Rumelhart and J. L. McClelland, Eds., Cambridge, MA, MIT Press, pp 318–362, 1986.

14. HONIK, K. M., STINCHCOCOMBE, M., AND WHITE, H., Multilayer Feedforward Networks are Universal Approximators, *Neural Networks, Volume 2*, pp 359–366, 1989.

15. ZADEH, L. A., Fuzzy Sets, *Information and Control, Volume 8*, pp 338–353, 1965.

16. CHEN, S.-M., Weighted Fuzzy Reasoning Using Weighted Fuzzy Petri Nets, *IEEE Transactions on Knowledge and Data Engineering, Volume 14*, no. 2, pp 386–387, 2002.

17. BURROUGH, P. A., Fuzzy and Fractal Objects for Intelligent Spatial Analysis in Geographical Information Systems, in *Modelling and New Methods in Image Processing and in Geographic Information Systems*, P. Mandl, Eds., Tome 61, R. Oldenbourg, Vienna - Munich, pp 37–38, 1992.

18. KOSCO, B., *Neural Networks and Fuzzy Systems: A Dynamic Approach to Machine Intelligence*, Prentice Hall Incorporation, Englewood Cliffs, NJ, USA, pp 139–149, 1992.

19. PETERSON, J. L., Petri Nets Theory and Modeling of Systems, Prentice Hall, Englewood Cliffs, N.J., USA, 1981.

20. CHEN, S.-M., Ke, J.-S., and Chang, J.-F., Knowledge Representation Using Fuzzy Petri Nets, *IEEE Transactions on Knowledge and Data Engineering, Volume 2*, no. 3, pp 311–319, 1990.

21. XINGUI, H., Weighted Fuzzy Logics and Its Applications, in *Proceedings of Twelfth Annual International Computer Software and Application Conference*, pp 485–489, 1988.

22. RASMANI, K. A., AND SHEN, Q., Modifying Weighted Fuzzy Subsethood-based Rule Models with Fuzzy Quantifiers, *Technical Report*, The University of Edinburgh, UK, pp 79–84, 2004.

23. TSEKOURAS, G. E., On the Use of the Weighted Fuzzy C-means in Fuzzy Modeling, *Advances in Engineering Software, Volume 36*, pp 287–300, 2005.

24. HAGRAS, H., CALLAGHAN, V., AND COLLEY, M., Online Learning of Fuzzy Behaviours using Genetic Algorithms and Real-time Interaction with the Environment, in *Proceedings IEEE International Fuzzy Systems Conference*, Seoul, Korea, Volume II, pp 668–673, 1999.

25. WSM, *World Shipyard Monitor, Volume 13*, no. 1, Jan. 2006, 1–107, 2006.

26. KGSE, *KG Shipping and Finance 2006*, German KG System, Germany, pp 1 - 45, 2007.

27. SIW, *Shipping Intelligence Weekly*, Clarkson Research Services, UK, pp 1–37, 2007.

28. LNGTT, *LNG Trade and Transport 2006*, Clarkson Research Services, UK, pp 1–302, 2007.

29. CIW, *China Intelligence Weekly*, Clarkson Research Services, UK, pp 1–137, 2007.

30. ADB, *ADB Asian Development Outlook (2006–2007) - (2010–2011)*, Technical Report (TR-ADB:ADO/2006–11), Philippines, 2007.

31. UNR, *Regional Shipping and Port Development Strategies (Container Traffic Forecast)*, Monograph Series on Managing Globalization, Economic and Social Commission for Asia and the Pacific, UNTR - ST/ESCAP/2398, United Nations, New York, USA, pp 1–62, 2005.

32. DSC, *The Drewry Annual Container Market Review and Forecast 2005/06*, Drewry Shipping Consultants, London, UK, pp 1–23, 2005.

33. NOTTEBOOM, T. E., Container Shipping and Ports: An Overview, *Review of Network Economics, Volume 3*, no. 2, pp 7–18, 2004.

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